

1 CODE BASIS

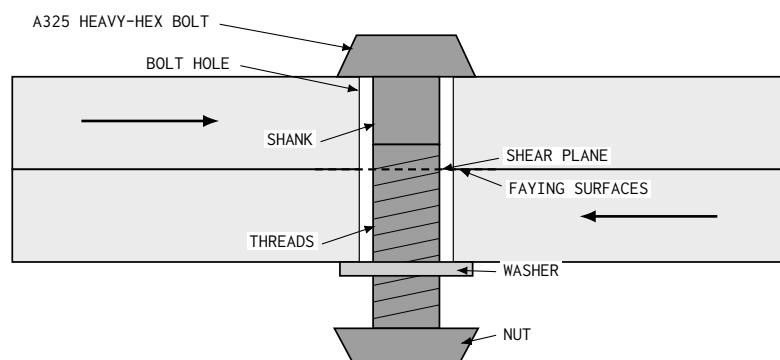
IBC 2018

ANSI/AISC 360-16, SECTION N5.6

RCSC 2014, SECTION 9.1

CURRENT: ASTM F3125 GRADE A325

2 CONNECTION USE



A325-N

BEARING-TYPE CONNECTION
THREADS INCLUDED IN SHEAR PLANE

A325-X → THREADS EXCLUDED FROM SHEAR PLANE

A325-SC → SLIP-CRITICAL CONNECTION

N, X, and SC describe connection use—not different bolt materials.

3 TYPICAL USES

- Beam shear connections.
- Beam and column splices.
- Gusset and bracing connections.
- End-plate connections.
- Steel-to-steel joints carrying shear or combined shear and tension.

5 NOT REQUIRED SOLELY BECAUSE BOLT IS A325-N

- Measurement of actual bolt clamping force.
- Generic torque-wrench calibration.
- AWS D1.1 certification for the bolt installer.

4 INSPECTION – TYPICAL SNUG-TIGHT A325-N JOINT

PRIOR TO BOLTING

- Confirm specified bolt, nut, and washer components.
- Confirm required ASTM markings, grade, type, diameter, and length.
- Confirm proper storage and suitability for installation.
- Confirm holes and connected steel plies meet joint requirements.

AFTER ASSEMBLY

- Confirm connected plies are in firm contact.
- Confirm washers are installed where required.
- Confirm every bolt is snug-tight and the nut cannot be removed without a wrench.
- Document acceptance or rejection of the connection.

PRETENSIONED OR SLIP-CRITICAL JOINTS REQUIRE ADDITIONAL, INSTALLATION-METHOD-SPECIFIC INSPECTION. A325-N ALONE DOES NOT REQUIRE THAT ADDITIONAL INSPECTION.